

Title: Innovation Hub

Fostering collaboration and innovation among Engineering students

Problem Statement

Students at Lupane State University often work on school projects in isolation, lacking opportunities for interdisciplinary collaboration, knowledge sharing and real-world problem solving. This can lead to lack of innovation, limited exposure to diverse perspectives and inadequate preparation for the industry. The current methods of collaboration such as social media groups and emails are often ineffective and disorganized

Introduction

Innovation hub is a web-based platform designed to bring together engineering students from diverse disciplines (Mechatronics, business computing and Information technology) to collaborate on projects, share ideas and develop innovative solutions. The platform aims to bridge the gap between students' academic learning and real-world problem solving, preparing them for the challenges of the industry.

Research objectives

1. Design and develop a user-friendly platform for students to create, share and collaborate on projects
2. Foster a community of innovators, encouraging interdisciplinary collaboration and knowledge sharing
3. Provide a platform for students to showcase their projects and achievements
4. Evaluate the effectiveness of Innovation Hub in promoting collaboration, innovation, and employability among engineering students

Benefits

1. Cross-disciplinary collaboration: Mechatronics, Business computing and Information technology students can work together fostering learning and innovation
2. Practical experience: Student gain hands on experience working on real projects, building portfolios and networks
3. Entrepreneurial opportunities: Potential for projects to become start up ideas or attract external funding

Core Features

1. Project creation: Users can create project ideas, add descriptions and specify required skills
2. Team building: Users can apply to join projects and project owners can review and approve/deny applications
3. Skills matching: Students can showcase skills and interests, making it easier for project owners to find the right team members
4. Project showcase: A directory of active projects, allowing students to browse and join projects that interest them
5. Discussion Forum: A space for team members to discuss project progress, ask questions and share resources
6. Project registration: Admins can register student projects, assigning a unique identifier and time stamp to prevent idea theft and ensure originality
7. Notification system

How this system would work?

1. Students sign-up and create profiles showcasing their skills and interests
2. Students submit project for registration
3. Users create projects ideas, specifying required skills and team size
4. Project owners can browse student profiles to find team members
5. Students can apply to join projects or create their own teams
6. Teams can collaborate, share resources and track progress within the platform

Methodology

1. Literature review
Conduct a comprehensive review of existing collaboration platforms, identifying best practices and areas for improvement
2. Requirements gathering
Conduct surveys and focus groups with engineering students to understand their needs, preferences and pain points
3. Platform Development
Design and develop the Innovation Hub using an iterative and agile approach, incorporating feedback from students and lecturers

Innovation Hub roadmap:

- Finalize features, goals and timelines
- Setting up the development environment (Choose your IDE, install necessary libraries and configure the database)
- Design the database schema
- Build core features (user authentication, project creation and team building)
- Test and iterate (testing each feature, gather feedback and refine the implementation)

Tools

- Backend
Python with flask or Django: For building the web app, handling requests and interacting with the database
- Frontend
HTML/CSS/JavaScript: For building user interface
- Database
PostgreSQL or SQLite: For storing user data, project information and team collaborations
- Real-time updates
Use of web sockets or library like Django channels for real-time updates and notifications

4. Testing and evaluation

Conduct usability testing (alpha and beta testing) and evaluate the platform's effectiveness through user feedback and metric analysis

Expected outcomes

1. A functional user-friendly platform for students to collaborate on projects
2. A community of innovators with increased interdisciplinary collaboration and knowledge sharing
3. Improved employability and entrepreneurial skills among the engineering students
4. A comprehensive evaluation of innovation hub's effectiveness, informing future development and implementation

Timeline

1. Literature review and requirements gathering 2 Weeks
2. Platform development 8 weeks
3. Testing and evaluation 4 weeks
4. Final report and presentation 2 weeks

Personnel

- Researcher: Itai Vera
- Supervisor:
- Participants:

Budget

Development costs

Web development tools and software: \$0.00 (using open source tools)

Domain registration \$100.00

Research costs

- Survey incentives
- Focus group refreshments: TBA

Miscellaneous

- Printing and materials \$20.00

Total costs: \$120.00

Resources

- Computing resources (laptop, software, internet)
- Incentives for participants (optional)

Thank You!!!

